COURSE SYLLABUS

ENGR 1105: Intro to Robotics & Mechatronics

Fall 2026

RANDOLPH COLLEGE

Instructor	Dr. Huan Phan, Assistant Professor of Robotics and Mechatronics Engineering
Email	hphan206@randolphcollege.edu
Class meetings	**Mon., Tue., Thu., Fri. • 9:30–10:45 a.m. • Martin Science Building, Room 208**
Course dates	October 26–December 14, 2026
Office hours	By appointment; weekly times posted on the course website
Course site	huantianh.github.io/courses/intro-robotics-mechatronics Announcements, readings, lab instructions, assignments, and schedule updates

Course Description

This hands-on course introduces robotics and mechatronics through mechatronic-system fundamentals, electrical circuits, Arduino programming, sensors and reliable measurement, Python data analysis, actuators and power systems, and final-project integration. Lectures directly prepare students for the corresponding laboratory work. Students also complete a separate, self-directed Mechanical Automata project and a final Controller-Based Arduino Project. No previous programming or electronics experience is required.

Course Learning Outcomes

By the end of the course, a successful student will be able to:

1. explain the roles of structures, mechanisms, sensors, controllers, actuators, and power sources in mechatronic and robotic systems;
2. construct, test, and troubleshoot basic breadboard and Arduino circuits using appropriate electrical quantities, polarity, and safe wiring practices;
3. write introductory Arduino programs using variables, conditionals, Boolean logic, Serial Monitor output, analog input, PWM, timing, and reusable functions;
4. characterize sensor behavior using range, resolution, accuracy, precision, calibration, thresholds, hysteresis, filtering, and repeated measurements;
5. collect, organize, plot, filter, calibrate, and interpret engineering sensor data using Python in Google Colab;
6. safely connect and control servos, DC motors, motor drivers, and external power sources while distinguishing logic signals from actuator power;
7. organize a controller using states, safe defaults, feedback or threshold logic, timeouts, and diagnostic information; and
8. integrate mechanical, electrical, sensing, programming, and actuation subsystems into a functioning Arduino-based project and demonstrate its reliability through systematic testing and documentation.

Course Materials

- Course readings, laboratory instructions, and project requirements posted on the course website.
- Arduino Uno-compatible starter kit and a laptop with the Arduino IDE.
- Google Colab for browser-based Python sensor-data analysis and visualization. A free Google account is required; no paid software license is required.
- Mechanical Automata and final-project materials approved by the instructor.
- An engineering notebook for sketches, diagrams, code notes, measurements, and test results.
- Safety glasses when instructed.

No textbook purchase is required unless announced by the instructor. Students should not purchase additional project components without instructor approval.

Course Organization and Expectations

The course meets four times per week for 75 minutes. Weeks 1–4 follow a two-lecture/two-lab rhythm. Lecture X.1 normally meets Monday, Lecture X.2 Tuesday, Lab X.1 Thursday, and Lab X.2 Friday. Week 5 is shortened because of Thanksgiving break and contains one actuator lecture and one shortened actuator laboratory. Week 6 focuses on final-project design, integration, state-based behavior, testing, reliability, and troubleshooting. Week 7 is reserved for final-project preparation, presentations, demonstrations, and the project showroom.

- **Project 1 – Mechanical Automata:** A separate, self-directed project completed outside the numbered laboratories. Students independently study a simple mechanism and build a non-motorized prototype that produces repeated animal-inspired motion. The instructor provides requirements, examples, checkpoints, and consultation.
- **Final Project – Controller-Based Arduino Project:** Build an approved Arduino-controlled system with a power source, at least one sensor or user input, at least one output or actuator, and repeatable programmed behavior. The final system should include safe startup behavior, documented subsystem interfaces, systematic testing, and evidence of reliable operation.

There are six numbered laboratory assignments. Each laboratory may span two class meetings but is submitted and graded as one assignment. Lab 5 is shortened because of Thanksgiving break, and Lab 6 serves as the structured final-project integration laboratory. There are no separate homework assignments. Students are expected to attend regularly, work safely, contribute to their team, and keep design notes, wiring records, code notes, measurements, and test results current.

Assessment and Grading

Assessment category	Weight
Labs (6)	30%
Project 1: Mechanical Automata	30%
Final Project: Controller-Based Arduino Project	30%
Peer Evaluation	10%
Total	**100%**

A	B	C	D	F
90–100%	80–89.9%	70–79.9%	60–69.9%	below 60%

Labs

The course includes six graded laboratory assignments. During a normal week, a laboratory may be divided into Lab X.1 and Lab X.2, but the two meetings are submitted and graded as one lab. Lab 5 is a shortened actuator laboratory, and Lab 6 is the final-project integration laboratory. Unless announced otherwise, the six labs contribute equally within the lab category. Lab grades reflect preparation, safe participation, circuit or system functionality, completion of required tests, troubleshooting, data or observations, and clear documentation and reflection.

Project 1: Mechanical Automata

The automaton must use no electric motor or electronic controller, produce repeated animal-inspired motion, and include at least one identifiable mechanism. Grading emphasizes functional motion, appropriate mechanism selection, design development, testing and improvement, documentation, and demonstration.

Final Project: Controller-Based Arduino Project

The final project must use an Arduino or compatible controller and include a power source, at least one sensor or user input, at least one output or actuator, and programmed decision logic. The project may be a robot, automated mechanism, interactive device, or another approved mechatronic system. Grading emphasizes safe construction, correct operation, subsystem integration, readable program organization, testing at normal and boundary conditions, reliable behavior, documentation, technical explanation, and final demonstration.

Peer Evaluation

Students confidentially evaluate teammates based on preparation, participation, communication, reliability, respect, safety, and contribution. Peer evaluations inform the individual peer-evaluation grade, but the instructor reviews all evaluations and determines the final score.

Laboratory Safety and Equipment Care

Follow all laboratory instructions and use tools and components only for their intended purpose. Wear safety glasses when instructed, secure long hair and loose clothing, disconnect power before changing wiring, and never power motors or other high-current devices directly from Arduino input/output pins. Return equipment in organized condition.

Course Policies

Attendance

Regular attendance is expected. In a seven-week course, one absence represents substantial material and project time. Notify the instructor as early as possible and remain responsible for missed work.

Late and Make-Up Work

Unless stated otherwise, late work loses 10 percentage points per calendar day. Contact the instructor before the deadline when illness or an emergency prevents timely completion. Missed laboratory work may require an alternative assignment.

Teamwork and Individual Responsibility

Students may collaborate on team projects, but each student must understand the team's design, wiring, code, and test results. Individual notebook entries and reflections must be the student's own work.

Communication and Conduct

Use Randolph College email and include ENGR 1105 in the subject line. Contribute to a respectful, inclusive, and productive learning environment. Phones and unrelated devices should be put away; laptops may be used for course work.

Randolph College Policies and Resources

Accommodations

Students seeking accommodations should contact Larvail Jones, Coordinator of Access Services, at ljones@randolphcollege.edu or 434-947-8132, then meet privately with the instructor after receiving an accommodation letter.

Title IX and Sexual Misconduct

Faculty members are mandatory reporters. Reports involving sexual misconduct or a campus crime must be shared with the Title IX Coordinator, Jaclyn Beard, at jbeard@randolphcollege.edu or 434-947-8031. The Counseling Center is a confidential resource at counselingcenter@randolphcollege.edu or 434-947-8130.

Honor Code

Students must submit their own work, identify collaborators, cite sources and software, and never invent measurements or test results. Violations will be handled under the Randolph College Honor Code. See `https://www.randolphcollege.edu/studenthandbook/`.

Artificial Intelligence

AI tools may be used for brainstorming, grammar, conceptual explanations, or debugging questions. They may not generate submitted designs, code, diagrams, reports, notebook entries, or reflections unless explicitly authorized. Permitted use must be disclosed, and students remain responsible for accuracy, safety, and originality.

Tentative Course Schedule

The schedule below reflects the lecture and laboratory materials developed for Weeks 1–6. Weeks 1–4 normally contain two lecture meetings and two laboratory meetings. Week 5 is shortened because of Thanksgiving break. Week 6 is devoted to project design and integration. Week 7 is reserved for final-project preparation, presentations, demonstrations, and the project showroom. Project 1 is completed separately from the numbered laboratories.

Week	Date	Topic or activity
1	Mon., Oct. 26	**Lecture 1.1 – Robotics and Mechatronics Foundations:** Mechatronic systems, real examples, the sense–decide–act model, major subsystems, Arduino overview, and Project 1 introduction.
1	Tue., Oct. 27	**Lecture 1.2 – Electrical and Arduino Foundations:** Voltage, current, resistance, power, Ohm's law, polarity, complete circuits, breadboards, circuit diagrams, digital output, digital input, safe wiring, and troubleshooting.
1	Thu., Oct. 29	**Lab 1.1 – Arduino Foundations:** Connect the Arduino, modify Blink, build an external LED circuit, and program a digital output.
1	Fri., Oct. 30	**Lab 1.2 – Arduino Foundations:** Wire and read a push button, create simple input/output behavior, troubleshoot the circuit, complete a design challenge, and submit Lab 1. Project 1 concept due.
2	Mon., Nov. 2	**Lecture 2.1 – Arduino Programming Foundations:** Program sequence, variables, constants, assignments, conditions, Boolean logic, program tracing, Serial Monitor output, timing, and controlled debugging.
2	Tue., Nov. 3	**Lecture 2.2 – Analog Input and PWM:** Digital versus analog signals, potentiometers, Arduino analog input, voltage conversion, PWM, LED brightness, threshold behavior, calibration, and measurement variation.
2	Thu., Nov. 5	**Lab 2.1 – Potentiometer and Analog Measurement:** Read raw analog values, convert readings to voltage, use Serial Monitor, and observe measurement variation.
2	Fri., Nov. 6	**Lab 2.2 – PWM and Threshold Behavior:** Add a PWM LED circuit, control brightness from analog input, create threshold-based behavior, troubleshoot, complete a design challenge, and submit Lab 2.
3	Mon., Nov. 9	**Lecture 3.1 – Sensors and Reliable Measurement:** Sensor types, the measurement chain, analog and digital sensing, range, resolution, accuracy, precision, calibration, photoresistors, thresholds, and hysteresis.
3	Tue., Nov. 10	**Lecture 3.2 – Ultrasonic Distance and Reliable Decisions:** Trigger–echo timing, distance calculation, invalid readings, noise, outliers, moving averages, warning zones, hysteresis, repeated confirmation, and troubleshooting.
3	Thu., Nov. 12	**Lab 3.1 – Photoresistor Calibration and Thresholds:** Build the light-sensor circuit, collect measurements, choose a threshold, control an LED, and add hysteresis.
3	Fri., Nov. 13	**Lab 3.2 – Ultrasonic Distance and Reliable Decisions:** Measure distance, test unreliable targets, average valid readings, build distance-warning zones, complete a design challenge, and submit Lab 3. Project 1 progress check.
4	Mon., Nov. 16	**Lecture 4.1 – Python Foundations and Engineering Plots:** Google Colab, variables, lists, NumPy arrays, zero-based indexing, vectorized operations, pandas DataFrames, Matplotlib plots, decisions, loops, and simple functions.
4	Tue., Nov. 17	**Lecture 4.2 – Python Sensor Data Analysis and Reliable Decisions:** Import CSV data, validate measurements, filter with moving averages, calculate statistics, perform linear calibration, evaluate error, classify sensor states, and communicate results.

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Week	Date	Topic or activity
4	Thu., Nov. 19	**Lab 4.1 – Python Foundations:** Use Google Colab, create NumPy arrays, perform engineering calculations, import CSV data, and generate correctly labeled plots.
4	Fri., Nov. 20	**Lab 4.2 – Python Sensor Analysis:** Analyze Week 3 sensor data in Python, filter measurements, create a calibration model, evaluate error, define decision thresholds, interpret results, and submit the completed Colab notebook. Project 1 demonstration or video due.
5	Mon., Nov. 23	**Lecture 5.1 – Actuators, Motors, and Power Systems:** DC, servo, and stepper motors; torque and speed; starting and stall current; transistors; flyback protection; motor drivers; H-bridges; PWM; power sources; common ground; safe startup; and troubleshooting.
5	Tue., Nov. 24	**Lab 5 – Actuator Control and Safe Power Systems:** Complete the common power-and-safety checks, then use the assigned servo or DC-motor-driver track to test safe motion, user-input control, improved motion behavior, and a design challenge.
5	Thu., Nov. 26	**No class – Thanksgiving break.**
5	Fri., Nov. 27	**No class – Thanksgiving break.**
6	Mon., Nov. 30	**Lecture 6.1 – Project Design and Subsystem Integration:** Project purpose, measurable success criteria, constraints, block diagrams, feedback, subsystem interfaces, I/O and pin budgets, power budgets, mechanical layout, wire management, and incremental bring-up.
6	Tue., Dec. 1	**Lecture 6.2 – State-Based Behavior, Reliability, and Testing:** Open-loop and feedback behavior, state logic, thresholds, hysteresis, debouncing, timeouts, repeated confirmation, nonblocking timing, safe defaults, test matrices, troubleshooting, diagnostics, reliability, and documentation.
6	Thu., Dec. 3	**Lab 6.1 – Final-Project Planning and Subsystem Assembly:** Define success criteria, create the block diagram, interface table, pin and power plans, mechanical layout, and separately test mechanical, controller, input, and actuator subsystems.
6	Fri., Dec. 4	**Lab 6.2 – Programmed Behavior, Integration, and Testing:** Define operating states and safe defaults, organize Arduino functions, improve decision reliability, integrate subsystems incrementally, execute test and boundary matrices, troubleshoot, document design changes, and demonstrate readiness.
7	Mon., Dec. 7	**Final Project Preparation:** Complete integration, testing, troubleshooting, documentation, display materials, and presentation rehearsal.
7	Tue., Dec. 8	**Technical Presentations:** Teams explain project goals, requirements, system architecture, wiring, program behavior, testing, design changes, and results.
7	Thu., Dec. 10	**Showroom Setup and Practice:** Prepare displays, rehearse demonstrations, complete peer feedback, and make final corrections.
7	Fri., Dec. 11	**Final Project Showroom:** Public demonstrations and final-project evaluation.
7	Mon., Dec. 14	**Course Completion:** Final demonstrations if needed, final-project submission, individual reflection, peer evaluation, course wrap-up, and equipment cleanup.

The instructor may revise this syllabus when necessary. Substantive changes will be announced in class and posted on the course website.